

Aadhitya Balaji

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EDUCATION

University of Wisconsin-Madison

Madison, WI

B.S. in Computer Science and Data Science, Minor in Business

Expected May 2029

- GPA: 3.92/4.0 | Dean's List: Fall '25, Spring '26

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, R, SQL, HTML/CSS

Machine Learning & Data Science: PyTorch, TensorFlow, OpenCV, Pandas, NumPy

Cloud & DevOps: AWS (S3, Lambda, EC2, ECR, VPC), Docker, Git

Databases, Web Development & Tools: Postgres, VS Code, React, Firebase

AI Skills: Claude Code, GitHub Copilot, Cursor, MCP

TECHNICAL PROJECTS AND RESEARCH

PickleIQ (AI Pickleball Coach)

April 2026 – Present

Lead Software & Cloud Engineer

Early Venture

- Partnered with a commercial pickleball facility and built a personalized coaching application that extracts biomechanical metrics and generates player-specific recommendations via the Claude API
- Cut video processing time 3x by containerizing the pipeline with Docker, storing images in ECR, and running automated ingestion and inference via AWS Lambda and EC2
- Improved rally detection accuracy 20% by fitting gravity-parabola trajectories to 2D ball tracks, recovering true court position via 3D physics-based reconstruction
- Boosted ball-detection coverage 68x (0.5% → 34% of frames) by fine-tuning TrackNetV2 on a custom human-labeled ball-position dataset

UW-Madison Computational Biology and Machine Learning Lab

September 2025 – Present

Undergraduate Researcher

Madison, WI

- Designed MORFEA, an unsupervised CNN-LSTM autoencoder modeling embryo development as latent trajectories, training on 700+ time-lapse sequences for label-free embryo quality classification
- Built an initial feed-forward classifier on MORFEA's latent embryo embeddings as a baseline for predicting developmental stage (cleavage, morula, blastocyst), informing later architecture decisions
- Co-authored a manuscript on MORFEA, invited for submission into a Frontiers Research Topic on deep learning-driven medical imaging

Verdict

May 2026

Developer

Personal Project

- Built a full-stack adversarial AI application in Next.js where three Claude agents debate any decision, idea, or claim across three rounds before a judge delivers a verdict to provide sharper feedback than a single LLM
- Designed a streaming orchestration layer coordinating 7 sequential Claude API calls per session, using a custom NDJSON protocol to stream token-level output live to the client

PROFESSIONAL AND LEADERSHIP EXPERIENCE

Wisconsin Consulting Club

February 2026 – Present

Analyst | YouTube Music x Wisconsin Consulting Club Team

Madison, WI

- Conducted a deep-dive analysis of 5 competitor platforms, pulling usage metrics and visualizing conversion funnels to surface strategic gaps to improve user retention and increase the customer base
- Designed mockups for 2 feature deep-dives and created a user journey map, translating research into 8 feature recommendations that shaped 2 of the team's 4 final strategic deliverables

Leadership Initiatives

July 2023 – May 2025

Public Health Staff Intern

Washington, D.C.

- Directed 9 cross-functional intern teams consisting of up to 63 total researchers to deploy 11 targeted public health campaigns across workshops in Bauchi State, Nigeria